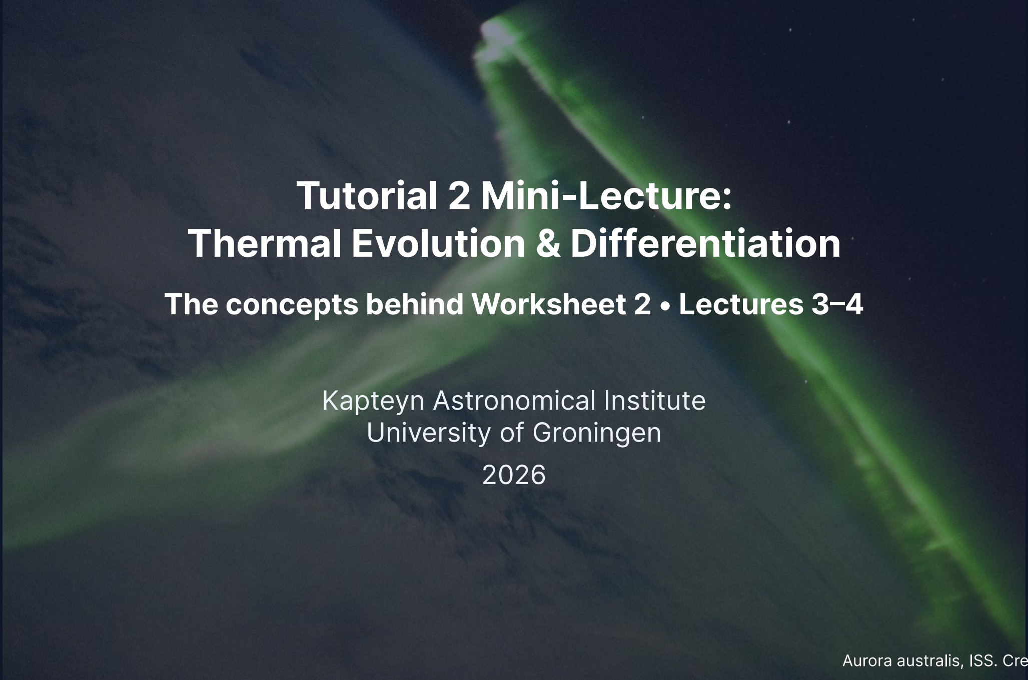
Tutorial 2 Mini-Lecture:
Thermal Evolution & Differentiation

The concepts behind Worksheet 2 • Lectures 3–4

Kapteyn Astronomical Institute
University of Groningen

2026



Aurora australis, ISS. Credit: NASA

What Worksheet 2 trains

Five problems, all built on Lectures 3–4. The physics you need, per problem:

1. **The radiogenic clock**: isotope table to heating power, and the two heating eras.
2. **Conduction fails, convection takes over**: $\tau = L^2/\kappa$, Ra, the $\Delta T^{4/3}$ flux law.
3. **A minimal thermal evolution model**: heat in minus heat out, on paper and from its solution curve.
4. **Building a core and dating it**: partitioning, the late veneer, Hf-W ages.
5. **Dynamos and shields**: standoff distance, ohmic decay.

Every problem has the shape and level of one exam question. No part needs a computer.

The radiogenic clock

Specific heating = concentration × heat per kg of isotope, summed:

$$h = \sum_i c_i h_i, \qquad \boxed{c_i(\text{past}) = c_i\, 2^{t/t_{1/2}}}$$

- Today: $\approx 4.9 \times 10^{-12}$ W kg^{-1} of BSE rock ⇒ ≈ 20 TW.
- The same $2^{t/t_{1/2}}$ law runs every clock: at 10 Myr, ^{26}Al is 13.9 half-lives down, a factor 1.5×10^4.

Two clocks, two eras

^{26}Al ($t_{1/2}$ = 0.72 Myr) out-heats everything for the first few Myr, then dies: planetesimals melt on the short clock, planets stay warm on the long one.

The isotope table is the whole calculation: concentrations, half-lives, heat rates. Everything else is bookkeeping.

Conduction fails, convection takes over

Conductive cooling time and convective vigour:

$$\tau = \frac{L^2}{\kappa}, \qquad \boxed{\mathrm{Ra} = \frac{\alpha \rho g \Delta T\, d^3}{\kappa \eta}}$$

- Moon: $\tau \approx 10^{11}$ yr $\gg$ age.
- Only bodies $\lesssim$ 400 km cool conductively in 4.57 Gyr.
- Mantle: $\mathrm{Ra} \approx 5 \times 10^7 \gg \mathrm{Ra}_c \approx 1708$.

How much faster is convection?

The Nusselt number $\mathrm{Nu} = (\mathrm{Ra}/\mathrm{Ra}_c)^{1/3}$ is the factor by which convection beats conduction. Since $\mathrm{Nu} \propto \Delta T^{1/3}$, the flux scales as $\Delta T^{4/3}$: hotter mantles shed heat disproportionately fast.

L^2 decides who conducts; Ra decides who convects; Nu says how vigorously.

A minimal thermal evolution model

One line of energy conservation:

$$\boxed{C\,\frac{\mathrm{d}T}{\mathrm{d}t} = H(t) - Q(T)}$$

- $H(t)$: radiogenic heating, a decaying exponential.
- $Q(T) \propto T^{4/3}$: convective loss (from $\mathrm{Nu} \propto \mathrm{Ra}^{1/3}$).
- You read peak, endpoint, and Urey ratio off the printed solution curve.

What the curve does

Start hot but under-radiating: the interior *warms* until Q catches H (~1 Gyr), then tracks the dying heat source down. Today: Urey ratio $H/Q \approx 0.4$.

Planets self-regulate: the $T^{4/3}$ thermostat ties the heat loss to the heat supply.

Building a core and dating it

One formula, one story:

$$\frac{C^{\mathrm{sil}}}{C^{\mathrm{bulk}}} = \frac{1}{1 + x(D - 1)}$$

- Siderophile elements ($D \gg 1$) vanish into the core.
- Their leftover excess in the mantle records the **late veneer**.

The Hf-W stopwatch

$^{182}\mathrm{Hf} \rightarrow {}^{182}\mathrm{W}$, $t_{1/2}$ = 8.9 Myr: Hf stays in the mantle, W leaves with the metal. The mantle's $\varepsilon^{182}\mathrm{W}$ excess dates core formation, until the parent goes extinct at ~45 Myr.

Core formation is chemically selective and datable: the mantle keeps both the receipt and the clock.

Dynamos and shields

The shield and its lifetime:

$$\boxed{r_{\mathrm{mp}} = R_p \left(\frac{B_0^2}{\mu_0 \rho_{\mathrm{sw}} v_{\mathrm{sw}}^2} \right)^{1/6}}, \quad \tau_{\mathrm{ohm}} = \frac{L^2}{\eta}$$

- For Ganymede's 300 km shell, τ_{ohm} ~ kyr.
- So an observed field is regenerated *now*: no fossil survives.

The sixth root

A tenfold solar-wind surge moves Earth's magnetopause only from 10 to 6.8 $R_{\oplus}$. Weak dependence is the shield's strength.

The pressure balance says how far the shield reaches; ohmic decay says a field must be driven today.

Working the worksheet

- **Show your algebra.** The exam asks for the steps, not only the number.
- Carry **4 significant figures** through intermediate steps; round at the end.
- Use the **checkpoints** (“show that …”) to verify your work and to keep moving if a step resists.
- Qualitative parts want **2 to 3 sentences** of physics, not an essay.
- Problem 3 prints the model’s solution curves: read the peak, the endpoint, and the Urey ratio straight off them.

Worksheet and full solutions: `ips.formingworlds.space/worksheets.html`.
We discuss questions, not presentations of the solutions.